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Certified Control Systems Technician® (CCST®)
Level I Study Guide

Setting the Standard for Automation™

The CCST® Study Guides are designed to provide a review of key content areas (domains) covered on the CCST examination and to familiarize the applicant with the type of questions that may be covered. Actual CCST examination questions are not available in the public domain and are not included in this study guide. Satisfactory completion of study guide questions in no way assures a passing score on the CCST examination.

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CCST Level I Study Guide

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Welcome

Thank you for your interest in the ISA Certified Control Systems Technician (CCST) Program. Being a CCST instantly makes you more valuable to your employer and increases your potential for higher profile jobs. The CCST Program was developed by leaders in the industry, including ISA, so your qualifications will be recognized industry-wide.

This self-study guide was developed to help you prepare for the ISA CCST Level I exam. You may have some idea of the structure of the CCST Program. If not, the CCST Introduction below will provide you with information about the Program.

This CCST Study Guide is divided into four domains to reflect the structure of the exam. Each domain consists of an introduction, sample questions, and other sources for study. Although the questions in this guide reflect questions on the exam, completing this guide is no guarantee of achieving certification. The best way to prepare for the exam is to study as much information relative to the domains and tasks as possible.

CCST Introduction

During early 1995, ISA, the International Brotherhood of Electrical Workers (IBEW), the Instrument Contracting and Engineering Association (ICEA), and the United Association of Plumbers and Pipe Fitters (UA) worked together to determine the needs for a control systems technician certification. After an extensive analysis of the market, ISA developed the CCST program.

CCST Program Overview

Control Systems Technician Definition

A Control Systems Technician (CST) is skilled in and knowledgeable of pneumatic, mechanical, and electronic instrumentation and understands process control loops and process control systems, including those that are computer based. Typically, a CST has received training in such specialized subjects as: theory of control, analog and/or digital electronics, microprocessors and/or computers, and the operation and maintenance of particular lines of field instrumentation. The CST is acquainted with both pipefitting and electrical fundamentals.

A CST performs calibration, documentation, loop checks, troubleshooting, and repair or replacement of instrumentation. These tasks relate to systems that measure and control level, temperature, pressure, flow, force, power, position, motion, physical properties, chemical composition, and other process variables.

Program Definition

ISA's CCST Program promotes the professional development of the CST, providing recognition and documentation of the technician's knowledge, experience, and education in measurement and control.

CSTs play a vital role in industry, exercising direct control over many operations that result in product quality, safety, and environmental protection. The CCST program ensures high levels of skill and competency, and is regarded as an important contribution by both industry management and labor.

To be certified under the ISA program as a CST, applicants must demon-

strate that they are skilled practitioners, knowledgeable in pneumatic and electronic instrumentation. They must demonstrate an understanding of process control loops and process control systems, including both analog and microprocessor based systems.

The CCST Program provides the mechanism to ascertain and document to management, contractors and labor organizations, the qualifications of CSTs to work safely and effectively in process industries worldwide.

As evidence of qualifying for certification, ISA will provide successful candidates with a certificate stating the level at which the individual is certified, and will maintain a registry of CCSTs who hold certificates.

Certification Classifications

CCST certifications are available at three levels, and each level requires the successful completion of an examination. Applicants qualify for a specific level based on combinations of experience and education.

CCST Program Requirements

Definitions

General criteria to qualify to take any of the following examinations include a commitment to the ISA Code of Ethics, a completed application with documentation included to support the level requirement, and the application fee.

Related experience always means work done in the instrumentation, measurement and control, electronics, electrical and mechanical fields. One (1) year of work experience means 1500 hours of active employment during a twelve (12) month period.

There is no academic requirement for any level. However, candidates may substitute post secondary education in related technologies for a maximum of four years work experience as defined for each level of certification. Related technologies include instrumentation, measurement and control, electrical, electronics or mechanical technologies. An academic degree, or equivalent, means an Associate of Science (AS) degree in these technologies or 40 semester hours in technology courses.

Documentation

Applicants must submit completed Verification of Employment forms for all employers or provide equivalent documentation that contains all information on the Verification of Employment forms in a clear and readable format. All forms must be signed by the supervisor. An applicant with military experience in the related technologies defined for the CCST program must document this period by submitting a copy of his/her DD 214 form accompanied by a written description of this experience. An official college transcript is also required to document educational periods needed to qualify for the program.

For employers that may be unavailable to provide first-hand verification, a two-party verification of the employment must be provided in the form of a statement signed by a co-worker from that period of employment and notarized by a notary public. A full explanation of why the employer is not verifying the period of employment must be provided. Full contact information, i.e., name, address and telephone number, must be

provided for all responsible parties who act to confirm employment verification.

No documents will be returned to applicants. Please keep a copy of your application for your records.

Record Retention Policy

All documentation received from CCST applicants who are not eligible for certification will be retained for one year after the application is received.

Education and Experience Requirements

Complete documentation of work experience and education are a **MUST** to fulfill the following requirements to qualify for each level of testing. Work experience and educational periods may not overlap when compiling the total number of years for a level.

Level I

A five-year total of education, training, and/or experience. The following elements may be used to satisfy this requirement:

a. A registered apprenticeship in an instrumentation, measurement and control, electronics, electrical, and/or mechanical program, if applicable.
b. Academic degree (or equivalent) in a related technology area, to a maximum of four years.
c. Successfully completing the CST Associate Recognition Program (equivalent to one year).
d. Related work experience.

Level II

A seven-year total of education, training, and/or experience. The following elements may be used to satisfy this requirement:

a. A registered apprenticeship in an instrumentation, measurement and control, electronics, electrical, and/or mechanical program, if applicable.
b. Academic degree (or equivalent) in a related technology area, to a maximum of four years.
c. Related work experience, including at least two years in instrumentation/measurement and control.

Level III

A thirteen-year total of education, training, and/or experience. The following elements may be used to satisfy this requirement:

a. A registered apprenticeship in an instrumentation, measurement and control, electronics, electrical, and/or mechanical program, if applicable.
b. Academic degree (or equivalent) in a related technology area, to a maximum of four years.
c. Related work experience, including at least five years in instrumentation/measurement and control.

The CCST program is a two-fold certification: (1) all applicants must meet the requirements for the level of testing for which he/she applies and (2) pass the examination. If proper documentation is not provided then the applicant will not be certified. In order to test again, the applicant must reapply and resubmit the application fee.

Eligibility

Once an application is received, candidates who are approved to take an exam are eligible for two testing windows , during a nine (9) month time period. If you do not test during this time period, you must re-apply to the program and submit a new application fee.

Examinations

A second comprehensive job analysis was conducted in 2001 to account for the work performed by CSTs. The job analysis identified performance domains that were subsequently validated in a nationwide survey for each of the three levels.

Level I and Level III Exams

The Level I and III exams are **closed book** exams of 175 multiple choice questions each; candidates are allowed four hours for completion.

Level II Exam

The Level II exam is a **closed book** exam consisting of eight written problems that test the candidate's ability to apply the knowledge and skills required for each domain in simulated practical situations. The exam assesses the candidate's ability to make decisions, troubleshoot, and make professional judgements in situations too complex or critical to test in a hands-on, practical examination; candidates are allowed four hours for completion.

Domains

The following table indicates the percentage of questions appearing on the exams as they relate to each domain. The percentage of questions per domain is based on each domain's relative importance and criticality in terms of what CSTs are expected to know while performing their jobs.

Domain		# of Questions		
		Level I	Level II	Level III
1	Calibration	31%	22%	19%
2	Loop Checking	21%	14%	13%
3	Troubleshooting	32%	22%	21%
4	Start Up	16%	12%	11%
5	Maintenance/Repair	NA	20%	20%
6	Project Organization	NA	10%	8%
7	Administration	NA	NA	8%

Taking the Exam

For more information regarding the CCST program, visit ISA online at www.isa.org/ccst, email ccst@isa.org, or call ISA at (919) 549-8411.

The examination is offered in an electronic format, or in the paper/pencil format for private exam sites. ISA may administer the CCST exam at special events and these may be found on the ISA Network. All candidates will have four hours to complete the exam.

Electronic exams are available during three testing windows as shown in the table below. For details on applying for electronic testing go to **www.isa.org/examschedule**.

Exam Testing Windows	Application Postmark Deadlines
Window 2: 1 July – 31 August	Thursday, 15 May
Window 3: 1 November – 31 December	Monday, 15 September
Window 1 2009: 1 March – 30 April	Thursday, 15 January

If the number of examinees for a private exam site or a special administration of CCST exams falls below the minimum

number pre-established for the session, ISA reserves the right to cancel the session and reschedule the affected examinees as quickly as possible. If an exam site must be cancelled because of a weather condition or insurmountable problems in regard to exam availability, test room availability, or proctor availability, ISA will contact the examinees as quickly as possible and arrange a replacement test date. Although rare, the potential exists for a cancellation to occur as late as the actual day of the exam.

All CCST exams are closed book. No reference material of any kind may be brought in the exam room.

Renewal

Renewal is required every three years. Renewal applicants must pay the renewal fee and submit a Renewal Application defining the Professional Development Points (PDPs) earned in the three year renewal period. 90 PDPs are required for renewal.

Once a candidate is certified, the Renewal Policies and Personal Log will be mailed with the CCST certificate and wallet card.

How Do I Renew?

If you renew on or before your certification expiration date, simply complete the Renewal Application and document 90 PDPs by:

- Documenting 3 years of work experience during the renewal period, OR
- Documenting a combination of work experience and continuing education, OR
- Registering for an examination.

Send your completed Renewal Application and $195 fee to ISA 60 days before your expiration date to keep your CCST active. If you renew after your certification expiration date but within the four month grace period, 120 PDPs will be required.

Disability

Applicants with disabilities as defined in Title III of the Americans With Disabilities Act who could be placed at a disadvantage when taking the CCST exam must advise ISA in writing of their needs by including a letter and necessary documentation, as appropriate, with the application material by the exam deadline stated on the exam schedule. ISA will contact the applicant to make the necessary arrangements before the exam date if the request is approved.

Fees

- The application fee to test at Level I is $295.
- The application fee to retest at Level I after failing the Level I exam is $170.
- The application fee for renewal by points is $195, and by exam is $295.

An applicant who wishes to retest must submit a Retest Form and test within their nine (9) month eligibility period to pay only the retest fee and avoid paying the entire application fee. An individual may only retest two times within the nine (9) month eligibility period. You must allow thirty (30) days between exam dates.

If you need to reschedule your electronic exam or private site exam, please refer to the CCST Program Handbook found at www.isa.org/CCST for details

on the reschedule fees. If you reschedule your exam with proper advance notice you will not incur reschedule fees. You may only reschedule one time, and the rescheduled exam date you choose must be within your nine (9) month eligibility period.

Fees are due in advance and ARE NOT REFUNDABLE. It is the responsibility of the applicant to thoroughly review the requirements for any examination and apply for the examination at the level at which he/she is qualified. No refunds will be made for applicants who do not show for testing on the scheduled exam date.

If a CCST misplaces or needs duplicate copies of the CCST certificate or wallet card, send a written request to ISA with payment of $15.00 per reprint of the certificate or wallet card ($30.00 for both).

Make checks payable to ISA. ISA does not issue receipts for payment of exam fees. Your canceled check is your proof of payment.

The application fees are subject to change. Examination fees may be tax deductible on your individual federal income tax return if your certification is to maintain or improve your professional skills. Payments for examination fees are not deductible as charitable contributions.

Completing the Application

Candidates must complete all sections of the application.

Section 1.

Provide complete contact information as requested. If ISA has a different mailing address on file for you, it will be changed to this Preferred Mailing Address. All ISA related materials will be mailed to this address unless you specify otherwise. Indicate if you suffer from a disability or religious obligation that would require an accommodation to test administration. Provide a full explanation of the accommodation you request.

Section 2.

Indicate the industry in which you currently work.

Section 3.

Indicate the level at which you are applying to sit for an examination. Please review the requirements for each level of examination. It is solely the responsibility of the applicant to request an exam at a level which he/she is qualified according to the experience and education requirements specified for that level. Applicants must provide proper verification for the level for which they apply. Applicants without required documentation will be notified to submit the necessary verification. If the applicant does not respond within six (6) months, the application will be considered abandoned. Applicant will have to submit a new application and the fee again.

Section 4.

Complete a record of employment that fully covers the experience requirement for the level at which you are applying to sit for an examination. Section 4 must be completed in addition to any other documentation submitted to verify employment.

Section 5.

Complete a record of education and training that fully covers the education requirement for the level at which you

are applying to sit for an exam. Official transcripts must be submitted as documentation to verify educational degrees. Certificates do not serve as verification of an apprenticeship or training. This experience must be verified by signed documentation.

Section 6.

Verification of Employment forms must be completed and signed for all work experience claimed as qualification for examination. All documentation must be provided in English. Faxed verification is not accepted. The original signature must be provided on this form.

Section 7.

Read carefully and sign the compliance agreement.

Section 8.

All applications must be accompanied by full payment of the required fees. No exceptions will be granted. **Purchase orders are not accepted.**

Section 9.

Mail application and fee to ISA. Incomplete or incorrectly completed applications may be returned to an applicant. It is the responsibility of the CCST applicant to promptly notify ISA in writing of any changes in his/her name or address. Notification for admission to the examination, test results, and renewal of certification depends on accurate information.

Submitting the Application

Once an application is submitted to ISA, the applicant must test within nine (9) months of the postmarked date. All examination materials must be sent to one of the following addresses:

1. If mailing check payment and/or mailing by regular postal delivery:
   ISA
   CCST Program
   P.O. Box 3561
   Durham, NC 27702

2. If mailing credit card payment and/or mailing by overnight service:
   ISA
   CCST Program
   67 Alexander Drive
   Research Triangle Park, NC 27709
   (919) 549-8411

Materials sent to any other address will be returned. **Fax, photocopied, and e-mail applications, or applications completed in pencil, are not accepted.**

Certification Status

Each applicant will receive an eligibility letter for electronic candidates or confirmation letter for private/special event exam candidates. For private/special event exams the confirmation letter will also include the time and location of the testing site.

Each applicant will be mailed notice of his/her certification status within thirty (30) days after the examination date. Electronic exam candidates will receive immediate results at the test center.

Due Process Procedures

If an applicant feels he/she was wrongly denied certification, original or renewal, from the CCST program, then he/she has the right to appeal.

I. Appeals
   A. All appeals shall be in writing.
      1. The notice of appeal must be delivered to the Certification Board, addressed to the attention of the ISA Executive Director, by the close of business on the 21st day after the date of the notice of denied certification.
      2. The appeal should include the date the notice of denied certification was received and must state the reasons the applicant believes the denied certification was in error.
      3. The appeal should indicate whether or not the applicant requests a hearing.
         a. If a hearing is requested, the applicant must explain why a hearing is needed, identify the issues to be resolved at a hearing, list names of prospective witnesses, and identify documentation and other evidence to be introduced at the hearing before the Board.
      4. The Chair of the Certification Board will select a three-member panel of the Board to become the Review Panel, one of whom shall be appointed Chair and will be the final vote in the event of a tie during the ruling.

II. Procedures
   A. The Review Panel will review the appeal and any request for a hearing. The Review Panel will grant a hearing in connection with the appeal, if requested.
      1. Hearing
         a. The Chair of the Review Panel will determine the time and location of the hearing within 90 days after determination that a hearing is warranted and will notify the applicant within the first 21 days. The applicant will be notified of the hearing time and location at least 20 days prior to the time determined for the hearing.
         a) The applicant may be represented by counsel or represent him/herself at the hearing. The applicant may offer witnesses and documents and may cross-examine any witness.
         b) The Review Panel may consider any evidence it deems relevant without regard to strict application of legal rules of evidence.
         c) The applicant is urged to submit a written brief (four copies) 10 days prior to the hearing to the Certification Board, addressed to the attention of the ISA Executive Director, for distribution to the panel in support of his/her position. However, written briefs are not required.
         d) Deposition
            i If the applicant or Review Panel desires to take a deposition prior to the hearing of any voluntary witnesses who cannot attend the hearing,

the deposition of a witness may be applied for in writing to the Chair of the Review Panel together with a written consent signed by the potential witness that he or she will give a deposition for one party and a statement to the effect that the witness cannot attend the hearing along with the reason for such unavailability.

ii The party seeking to take the deposition of a witness shall state in detail as to what the witness is expected to testify.

iii If the Chair of the Review Panel is satisfied that such deposition from a possible witness will be relevant to the issue in question before the Panel, then the Chair will authorize the taking of the deposition. The Chair will also designate a member of the Panel to be present at the deposition.

iv The deposition may be taken orally or by video. Any refusal of the taking of the deposition by the Chair shall be reviewed by the Panel at the request of the applicant.

v The party requesting the deposition will pay for the cost connected with taking the deposition.

B. The applicant will be notified of the result of the Review Panel within 30 days of the meeting.

II. Final Review

A. If the Review Panel makes a decision adverse to the applicant, the applicant may appeal to the full Certification Board for a final review by the close of business on the 21st day after the notice of decision is issued. Such appeal shall follow the same procedures as the initial appeal to the extent possible with the Chair of the Board serving as the Chair of the Final Review Panel. The Chair will once again select a three-member panel, one of whom shall be himself, to become the Review Panel, and the final vote in the event of a tie during the ruling shall be his.

B. If the applicant does not petition the Board for review or request a hearing before the Board regarding the recommendation of rejection of the application within the time allowed by these rules, the original decision by the Review Panel shall stand.

The ISA Code of Ethics

Preamble

As engineers, scientists, educators, technicians, sales representatives, and executives in an important and learned profession; and in order to safeguard public welfare; and to establish and maintain a high standard of integrity and practice; and as members of ISA, we hold to these Articles:

Article I

Members shall hold paramount the safety, health and welfare of the public in the performance of their duties, and shall notify their employer or client and such other authority as may be appropriate where such obligations are abused.

Members shall hold in confidence facts, data and information obtained in a professional capacity, unless the release thereof is authorized by their employer or client, and shall not engage in fraudulent or dishonest business or professional practices.

Article II

Members shall perform services only in areas in which they are qualified by education or experience, and shall endeavor to maintain their professional skills at the state of the art. Members shall practice their profession in a manner which will uphold public appreciation of the services they render.

Article III

Members shall issue public statements only in an objective and truthful manner, and shall include all pertinent and relevant information in professional reports, statements and testimony. Members shall be honest and realistic in making estimates or in stating claims based on available data. Members shall offer honest criticism of work, and shall properly credit the contributions of others.

Article IV

Members shall act in professional matters for each employer or client as faithful agents or trustees, and shall not participate in any business association, interest or circumstances which influence, or appear to influence, their judgment or the quality of their services. Members shall accept compensation, financial or otherwise, from only one party for services on or pertaining to the same work, unless otherwise agreed to by all parties; and shall not give or accept, directly or indirectly, any gift, payment or service of more than nominal value to or from those having business relationships with their employers or clients.

Article V

Members shall use only proper solicitation of employments, and shall represent their abilities, qualifications, education, technical associations and professional registrations without exaggeration and in accordance with the laws of the locations in which they practice.

Article VI

Members shall pledge themselves to live and work according to the laws of man and to the highest standards of professional conduct, using their knowledge and skills to the benefit of all mankind.

Resource Materials

The CCST program covers a broad area of domains in the field of automation and control. ISA has reference publications and training services to assist those preparing for certification and interested in skill enhancement. Study Guides are available for each level of certification. The Level I and III guides contain sample questions and listings of training products and courses related to each of the performance domains. The Level II guide allows you to familiarize yourself with the exam structure through a sample simulation problem.

ISA also offers training courses especially designed to assist those preparing for the CCST program. A list at the back of this guide provides more details. Any of these study products may be ordered by calling ISA at (919) 549-8411, or log on to the ISANetwork at **www.isa.org**.

The ISANetwork contains links to certification information, ISA's publications and training catalogs, and describes membership benefits and explains how to join ISA.

Why Should You Join ISA?

Founded in 1945, ISA (www.isa.org) is a leading, global, nonprofit organization that is setting the standard for automation by helping over 30,000 worldwide members and other professionals solve difficult technical problems, while enhancing their leadership and personal career capabilities. Based in Research Triangle Park, North Carolina, ISA develops standards; certifies industry professionals; provides education and training; publishes books and technical articles; and hosts the largest conference and exhibition for automation professionals in the Western Hemisphere.

Here are some of the benefits you enjoy as an ISA Member:

1. Stay current within your field by reading your free *InTech* magazine every month.

2. Make professional contacts and open the door to your future by participating in Technical Divisions and other networking activities, such as local Section meetings, conferences, and symposia.

3. Receive valuable money-saving discounts on all ISA training courses, books, videos, and CD-ROMs. You can also access free web seminars.

4. Shop smarter by using the *ISA Directory* — ISA's exclusive annual guide of products, specifications, manufacturers, sales offices, services, and more.

5. Keep up with industry standards and practices. As an ISA Member, you are able to download most ISA standards at no charge.

Get ahead by getting involved in ISA. To join ISA, complete an ISA Membership enrollment form and return it with your CCST Application, or call (919) 549-8411.

PERFORMANCE DOMAIN 1

CALIBRATION

Task 1 Obtain all documents needed for device calibration in order to proceed with calibration.

Task 2 Identify the correct device to be calibrated through documentation and instrument identification systems in order to ensure proper calibration.

Task 3 Select the correct test equipment through documentation and visual inspection as identified by site calibration procedures in order to correctly calibrate the device.

Task 4 Connect the test equipment to the device as required by manufacturer specifications and site procedures in order to properly and safely calibrate the device.

Task 5 Apply input values to the device as specified on instrument data sheet in order to simulate process parameters.

Task 6 Record "as found" readings on calibration sheets in order to provide a history of the device.

Task 7 Compare "as found" readings with the required values as specified by the instrument data sheet in order to identify if the device needs to be adjusted according to allowable tolerances.

Task 8 Make adjustments to the device, if necessary, through site calibration procedures in order to ensure accuracy.

Task 9 Complete all documentation as required in order to update all calibration records.

1. What pressure, in psi, would be indicated by an absolute pressure gage that is lying on a work bench not connected to anything?

   A. 0
   B. 29
   C. 14.7
   D. 2.9

2. A gage pressure of 195 psi is roughly what in psia?

   A. 151
   B. 164
   C. 178
   D. 210

3. Which of the following will have the highest pressure difference between the surface and 6 in. below the surface?

   A. a 12 in. vertical straw filled with water
   B. a 10 ft diameter open tank filled with water
   C. a 5 ft diameter vertical pipe filled with water
   D. none of the above, because the pressure difference in each is the same

4. If a cubic foot of water weighs 62.4 pounds in a 1 ft. cube, it exerts a pressure of how many pounds per square inch (psi) on the bottom of the container?

   A. 0.036
   B. 2.308
   C. 0.433
   D. 14.7

5. Hydrostatic head is another way of expressing:

   A. gallons.
   B. pressure.
   C. length.
   D. mass.

6. All physical measurements are based on the fundamental quantities of:

   A. mass, length, and time.
   B. density, viscosity, and specific gravity.
   C. flow, volume, and temperature.
   D. pressure, density, and specific gravity.

7. Gallons per minute is a unit of measurement for:

A. level.
B. flow.
C. pressure.
D. temperature.

8. Dividing the density of the process fluid by the density of ____ gives a dimensionless number called specific gravity.

A. mercury or mercury vapors
B. oil or natural gas
C. water or air
D. any known liquid or gas

9. As the speed (velocity) of a fluid increases through a restriction, the fluid pressure:

A. increases.
B. decreases.
C. remains unchanged.
D. first increases then decreases.

10. What Celsius temperature reading is within an oven temperature of 215 to 220°F?

A. 103
B. 110
C. 329
D. 338

11. The difference between the indication of the instrument and the actual value of the measured variable is:

A. the controlled variable.
B. called set point.
C. an error in the measuring instrument.
D. discarded.

12. Hydrostatic head, weight, radioactive properties, density, and sonic detectors are all used in what methods of level measurement?

A. point
B. pressure
C. inferential
D. direct

13. Differential pressure transmitters measure level by inference from:

A. capacitance.
B. conductivity.
C. hydrostatic head.
D. thermal energy.

14. Thermal elements infer level by measuring:

A. temperature.
B. capacitance.
C. thermal conductivity.
D. displacement.

15. All of the following types of level instruments can be used to detect the location of liquid surfaces except:

A. capacitance.
B. float.
C. hydrostatic head.
D. ultrasonic.

16. The temperature range of a transmitter is 100-200°F. The output signal range is 3-15 psig. What would be the expected output signal for an input temperature of 150°F?

A. 6 psig
B. 9 psig
C. 11.25 psig
D. 12 psig

17. Seventy-five percent output on a 4-20 milliamp transmitter is:

A. 12 milliamps.
B. 15 milliamps.
C. 16 milliamps.
D. 20 milliamps.

18. What is the designation used for a transducer that converts current to voltage?

A. E/I
B. I/P
C. P/I
D. I/E

19. An instrument gives the same indication each time it measures a variable under identical conditions. This defines:

A. static accuracy.
B. dynamic accuracy.
C. sensitivity.
D. repeatability.

20. The range of values over which an input signal to a transmitter may be changed upon reversal of direction with no observable change in output is called:

A. instrument accuracy.
B. reverse precision.
C. zero point.
D. dead band.

21. The steady-state gain of a standard 4-20 mA electronic hydrostatic head level transmitter with a measurement range of 50-250 inches of water is:

A. 0.08 mA/inch of water.
B. 0.10 mA/inch of water.
C. 16 mA/200 inches of water.
D. 20 mA/250 inches of water.

22. The temperature range of a transmitter is 0-200°F. The output signal range is 3-15 psig. What would be the expected output signal for an input temperature of 150°F?

A. 6 psig
B. 9 psig
C. 11.25 psig
D. 12 psig

23. Using ANSI/ISA standards, the red lead from a thermocouple or extension wire is always the _____ lead.

A. positive
B. negative
C. neutral
D. ground

24. A measurement instrument error is the difference between the actual value of the measured variable and:

A. its true value.
B. the calibration standard.
C. input and output signals.
D. the indication of the instrument.

25. To ensure the accuracy of a pneumatic controller throughout its range, you must precisely adjust all of the following EXCEPT:

A. supply pressure.
B. input.
C. span.
D. zero.

26. A device that converts alternating current to direct current is called a:

A. rectifier.
B. ratio station.
C. differential amplifier.
D. comparator.

27. A test instrument that displays a waveform is a(n):

A. function generator.
B. ammeter.
C. multimeter.
D. oscilloscope.

28. If an open tube manometer indicates 6.11 in. of HG, what is the pressure measurement?

A. 2 psi
B. 3 psia
C. 17.89 psig
D. 3 psig

29. Which of the following could be used to calibrate a pressure gage 0 to 800 PSI?

A. dead weight tester
B. a mercury manometer
C. the plant instrument air supply
D. multi-meter

30. 2 psi is equal to _____ inches of water.

A. 1
B. 27.7
C. 14.7
D. 55.4

31. Which of the following is usually adjusted first when calibrating an electronic pressure transmitter?

A. linearity
B. span
C. zero
D. range

32. The calibration plot (Fig. 1-1, page 21) shows that the transmitter does need recalibration to correct for:

A. a zero error.
B. a span error.
C. both a zero and span error.
D. a linearity error.

33. The calibration plot (Fig. 1-2, page 21) indicates that the pneumatic controller has _____ and must be calibrated.

A. a zero error
B. a span error
C. both a zero and span error
D. a linearity error

34. This electronic DP cell (Fig. 1-3, page 22) requires calibration to correct for the:

A. zero error.
B. span error.
C. zero, span and linearity errors.
D. zero and span errors.

35. This pneumatic level transmitter (Fig. 1-4, page 22) has:

A. hysteresis.
B. a zero error.
C. a span error.
D. zero, span and linearity errors.

36. _____ is a relative measure of the time delay between two events, states or, actions.

A. Backlash
B. Dead Time
C. Hysteresis
D. Dead Band

CALIBRATION ERRORS

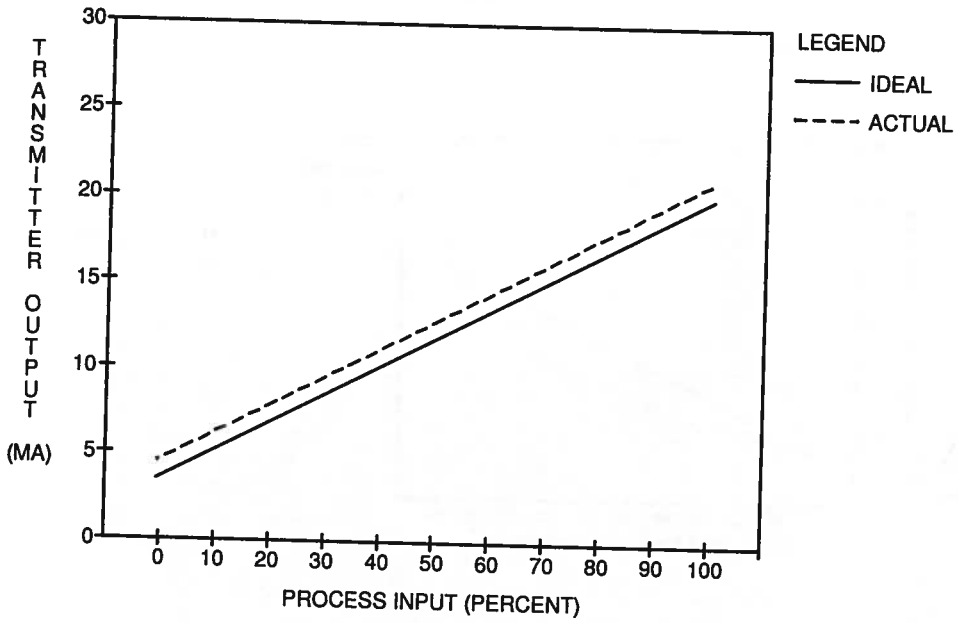


Figure 1-1

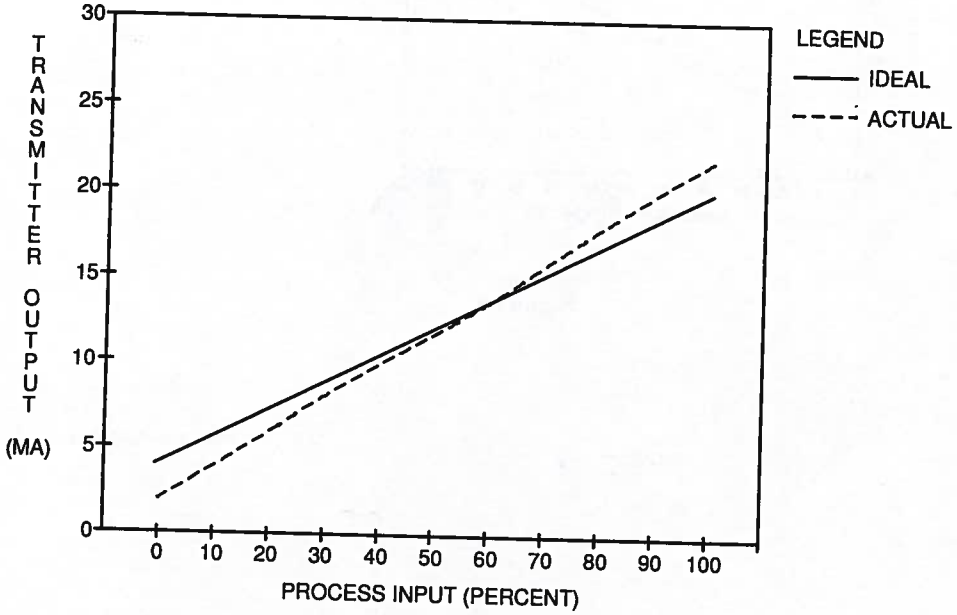


Figure 1-2

CCST Study Guide – Level I

22

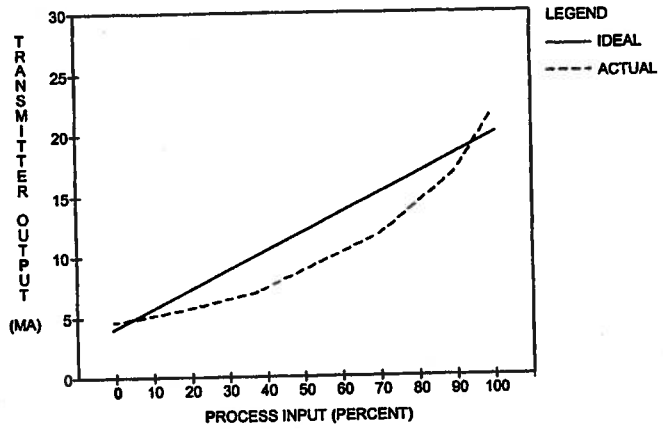


Figure 1-3

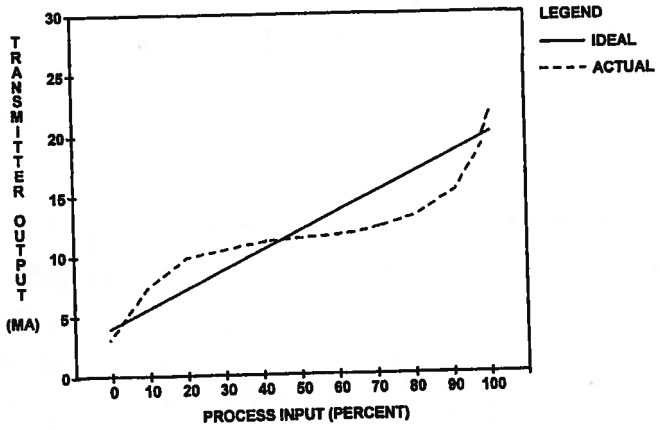
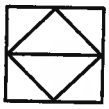


Figure 1-4

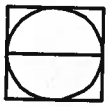
Domain 1 – Calibration

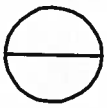
37. This function symbol from ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification* indicates a _____ in a primary location, normally accessible to the operator.



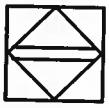
A. discrete instruments
B. shared display, shared control
C. programmable logic controller
D. a computer function

38. Which of the following symbols from ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification* is used to indicate shared display, shared control in a primary location normally accessible to the operator?

A. 

B. 

C. 

D. 

39. This primary element symbol  from ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification* identifies a(an):

A. Orifice plate.
B. venturi tube.
C. vortex sensor.
D. flow nozzle.

40. The Reynolds No. determines:

A. the venturi effect.
B. the coriolis acceleration.
C. the fluid viscosity.
D. if the flow is laminar or turbulent.

41. The flow rate in gallons per minute of water with a specific gravity of 1 provided by a pressure differential of 1.0 psi through a fully opened control valve is a definition of:

A. C_v.
B. DP.
C. O_v.
D. SG.

42. One significant advantage of an electromechanical actuator is:

A. low cost compared to pneumatic actuators.
B. easily made "fail safe (open or closed)."
C. normally, very low maintenance requirements.
D. it does not require clean, dry instrument air.

43. A displacer has a diameter of 4 in. and a length of 30 inches. If it is submerged to a depth of 20 in. in a liquid with a specific gravity of 0.8, how many pounds of upward buoyancy force will be exerted on the displacer? (round all digits to the hundredths place)

A. 9.07
B. 7.238
C. 35.5
D. 426.3

44. A plant has a 30 ft high water tank mounted on top of a 70 ft platform. What is the water level in the tank, measured in feet, if a pressure gage on the first floor, height 5 ft, reads 40 psi?

A. 22.3
B. 27.3
C. 30
D. 10

45. A 0-30 psig gage is being calibrated. Starting at 0 psi, the technician applies a pressure of 15 psi. The gage reads 14.9 psi. Then the technician applies 30 psi. When the pressure is reduced to 15 psi, the reading is 15.2 psig. What type of error might these readings indicate?

A. hysteresis error
B. span error
C. zero shift error
D. linearity error

46. If the range of a 3-15 psi pneumatic transmitter is 50-350°C, what is the output signal value, in psi, at 198°C?

A. 5.7
B. 7.2
C. 8.9
D. 12.6

47. Memory is normally organized in bit words called:

A. bytes
B. instructions
C. nibblers
D. registers

48. In a test of the accuracy of a pneumatic comparator with a pressure gage, pressure readings of the test gage and the gage being tested are compared at a number of test points. Test point readings are made:

A. only as the pressure is slowly increased.
B. only as the pressure is slowly decreased.
C. in no particular sequence.
D. as the pressure is increased and as the pressure is decreased.

49. A properly calibrated pressure transmitter connected to a 20-ft tall tank with a water level of 15 ft would indicate a pressure of approximately:

A. 19 psig.
B. 10 psig.
C. 180 inches of water.
D. 150 inches of water.

50. The span, in inches of water, of a differential pressure transmitter used to measure the full level of a tank that is 5-ft, 6-in. high and contains a process material with a specific gravity of 0.9 should be:

A. 73.3.
B. 59.4.
C. 66.0.
D. 5.6.

51. Closing high and low pressure tap block valves, venting the low side to atmosphere and connecting and applying a known calibration pressure to the high side is done to check:

A. the condition of the block valves.
B. tubing for leaks.
C. zero and span of a dP cell.
D. none of the above

52. An oscilloscope waveform spans 5.5 vertical divisions. The vertical gain control is set at 2 volts/division. The approximate peak-to-peak voltage of the signal represented on the oscilloscope is:

A. 33 volts.
B. 22 volts.
C. 11 volts.
D. 2.75 volts.

53. When checking the calibration of an instrument, the first information you should record should be:

A. original span.
B. original zero.
C. calibration data before adjustment.
D. final calibration data.

54. What is the value of a resistor that is color coded, gray - black - orange - silver?

A. 51 K ohms + or - 10%
B. 80 K ohms + or - 10%
C. 51 ohms + or - 10%
D. 803 ohms + or - 10%

55. The time constant (first order lag) in a transmission system is the time necessary (after dead time) following an upset for the response curve to reach _____ of its final value.

A. 100%
B. 63.2%
C. 50%
D. 37.8%

56. Load cells and strain gages measure the amount of material in a tank by measuring:

A. thermal conductivity.
B. hydrostatic head.
C. ultrasonic frequency.
D. weight.

57. _____ is generally accepted as an eight-bit segment of a computer word.

A. Bus
B. Bit
C. Byte
D. Base

58. Which of the following is the most temporary form of information storage?

A. magnetic tape
B. EEPROM
C. hard disk
D. RAM

59. A hexidecimal value of 0D converts to _____ binary.

A. 00101101
B. 00011001
C. 00010001
D. 00001101

PERFORMANCE DOMAIN 2

LOOP CHECKING

Task 1 Obtain documents pertaining to the process loop in order to understand all devices and their intended functions.

Task 2 Inspect the loop components through visual observation in order to ensure that the components are correctly installed.

Task 3 Verify that appropriate utilities are available and operational through visual inspection in order to perform the loop check.

Task 4 Isolate the process loop from the system by following established site procedures in order to perform the loop check in a safe manner.

Task 5 Simulate a change in the measured variable in order to verify proper loop performance.

Task 6 Make necessary corrections in order to bring loop performance within specifications.

Task 7 Complete all documentation as required in order to update all loop records.

1. A typical process control loop component would be which of the following?

   A. sensor
   B. controller
   C. final control device
   D. all of the above

2. A diamond on an instrumentation diagram encloses information about:

   A. calibration.
   B. the set point.
   C. controller action.
   D. the power source.

3. Central connection points for signal lines are called:

   A. instrument ports.
   B. junction boxes.
   C. input/output junctions.
   D. field ports.

4. In a pneumatic loop diagram, tube and bundle numbers:

   A. appear in rectangles.
   B. are written on or near the signal line.
   C. appear in the notes section.
   D. are connected with a curved arrow to the signal line.

5. In an electronic loop diagram, junction box connections are represented by:

   A. circles.
   B. diamonds.
   C. ellipses.
   D. squares.

6. On an instrumentation diagram, the loop identification:

   A. consists of two, three, or four letters.
   B. is the same for all instruments in a specific loop.
   C. identifies the measured or initiating variable.
   D. appears in the upper portion of the instrument balloon.

7. Process instrument connection lines are indicated on an instrumentation diagram by:

A. bold solid lines.
B. fine solid lines.
C. bold broken lines.
D. fine broken lines.

8. The thermostat for a central air conditioning system is set at 68°F. This value is an example of the:

A. manipulated variable.
B. deviation.
C. set point.
D. controlled variable.

9. The statement; "Thermocouple extension wires must have the same thermoelectric properties as the thermocouple that they are used with," is:

A. true, if the extension wires are submerged in the process fluid.
B. true in all installations.
C. false. Extension wires must be made from either copper or brass.
D. false. The type of extension wire makes no difference.

10. To convert the output from a thermocouple to a standard 4 to 20 mA instrument signal, a(n) _____ with cold junction compensation would be used.

A. I/P Transducer
B. P/I Transducer
C. D/A Converter
D. mV/I Transducer

11. Iron constantan thermocouples are designated by the letter:

A. T.
B. J.
C. B.
D. S.

12. To change the 4 to 20 mA analog signal from a transmitter to the signal required by a digital controller, a(an) _____ must be a part of the instrument signal loop.

A. I/P Transducer
B. Signal Conditioner
C. D/P Transmitter
D. A/D Converter

13. A device that changes a voltage signal to a current signal is normally referred to as a(an):

A. P/I Transducer.
B. E/I Transducer.
C. D/P Transmitter.
D. V/C Converter.

14. A pneumatic control valve, without a positioner, is the final control element in a instrument loop. The transmitter and controller are both electronic. In order for this control system to operate, a(an) _____ must be installed in the instrument signal line between the controller and the control valve.

A. I/P Transducer
B. MV/I Transducer
C. D/A Converter
D. P/I Transducer

15. The transmission signals considered standard for most instrument systems are:

A. 4 to 20 psi and 3 to 15 mA.
B. 0 to 24 volts and 0 to 20 psig.
C. 3 to 15 psig and 4 to 20 mA.
D. 0 to 20 mA and 0 to 15 psig.

16. A process is _____ if heat is produced by a reaction.

A. endothermic
B. continuous
C. intermittent
D. exothermic

17. A process is _____ if heat input is required to produce a reaction.

A. endothermic
B. continuous
C. intermittent
D. exothermic

18. In most process control systems the final control element is a:

A. transmitter.
B. sensor.
C. controller.
D. valve.

19. A control valve with a direct acting actuator and is air to close will:

A. fail closed.
B. maintain position.
C. fail open.
D. fail indeterminate.

20. A pneumatic valve actuator that operates over an instrument signal range of 9-15 psi is referred to as:

A. split range.
B. fail safe.
C. air to close.
D. full range.

21. One significant advantage of single loop/stand alone digital controllers is:

A. flexibility of configuration.
B. 4 to 20 mA input and output capability.
C. internal I/P and P/I transducers.
D. immunity to electrical surges and lightning strikes.

22. Hydrostatic head level can NOT be measured with a:

A. dP cell.
B. resistance probe.
C. dip pipe.
D. pressure gage.

23. RTD is short for:

A. Reactive Tuning Deviation.
B. Resistance Temperature Detector.
C. Resistance Tone Dampening.
D. Resonant Temperature Device.

24. What is a device that changes a voltage signal to a current signal?

A. P/I Transducer
B. E/I Transducer
C. D/P Transmitter
D. V/C Converter

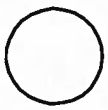
25. While increasing the pressure during a performance test, if you overshoot the specified test point by a small margin, you should:

A. take the reading at the pressure you have reached.
B. reduce pressure gradually to the test point value.
C. reduce pressure to zero and start again.
D. reduce pressure to below the test point and increase it again until the test point is reached.

26. The relationship between thermocouples and thermopiles is:

A. thermocouples consist of two or more thermopiles.
B. thermopiles consist of two or more thermocouples.
C. thermocouples only measure heat conductivity.
D. thermopile is another name for thermocouple.

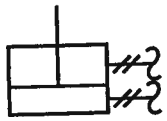
27. This function symbol from ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification* indicates a field mounted:

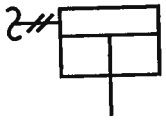


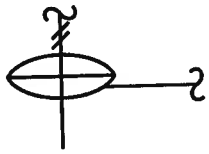
A. programmable logic controller.
B. computer function.
C. shared display, shared control.
D. discrete instrument.

28. Which of the following symbols from ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification* is used to indicate a pressure balanced diaphragm actuator?

A. 

B. 

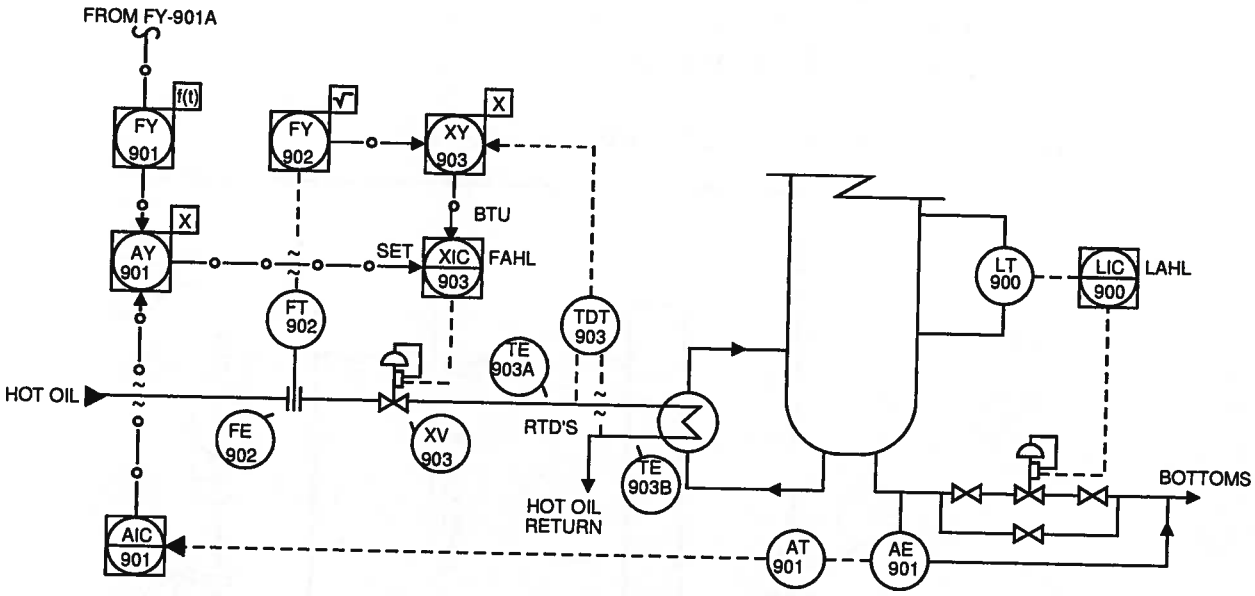
C. 

D. 

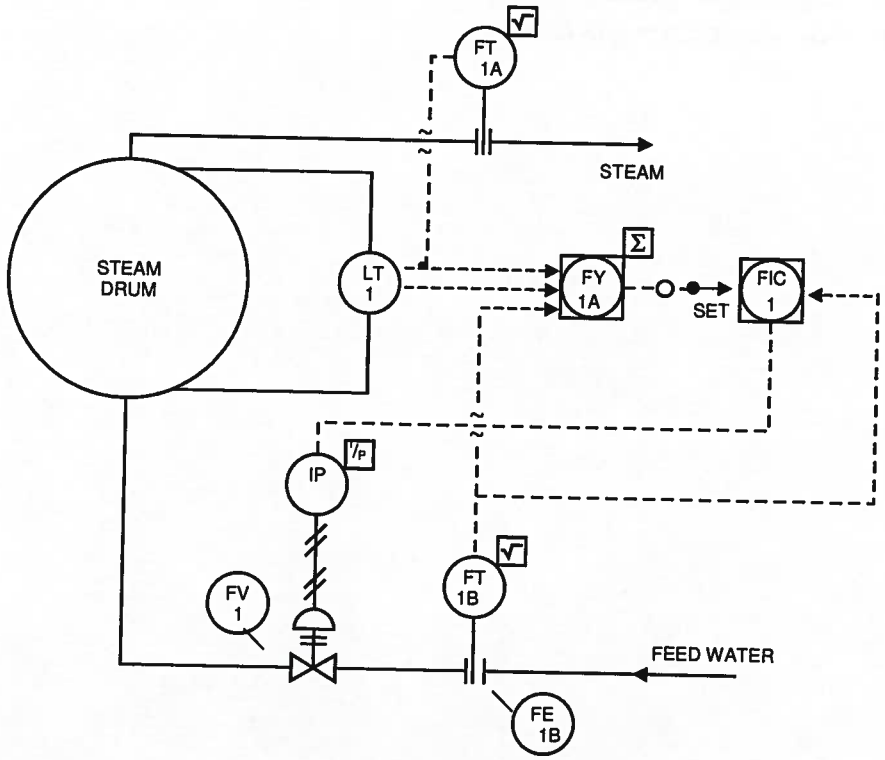
29. In the drawing (Fig. 2-1, page 35), LIC 900 is a(an):

A. shared flow computer with high/low alarms.
B. shared display indicating level controller.
C. discrete flow controller, field mounted.
D. pneumatic shared indicating controller.

Figure 2-1

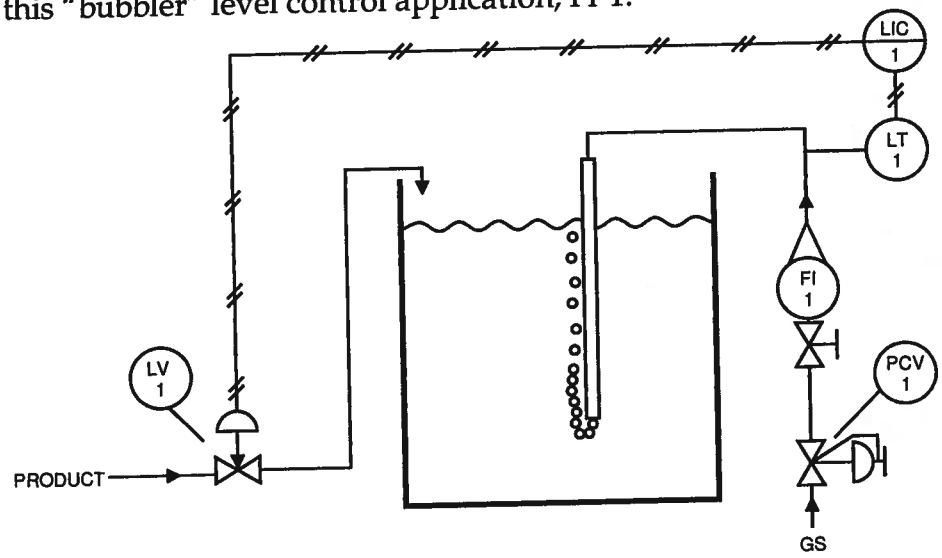


30. The control system for this steam drum is:



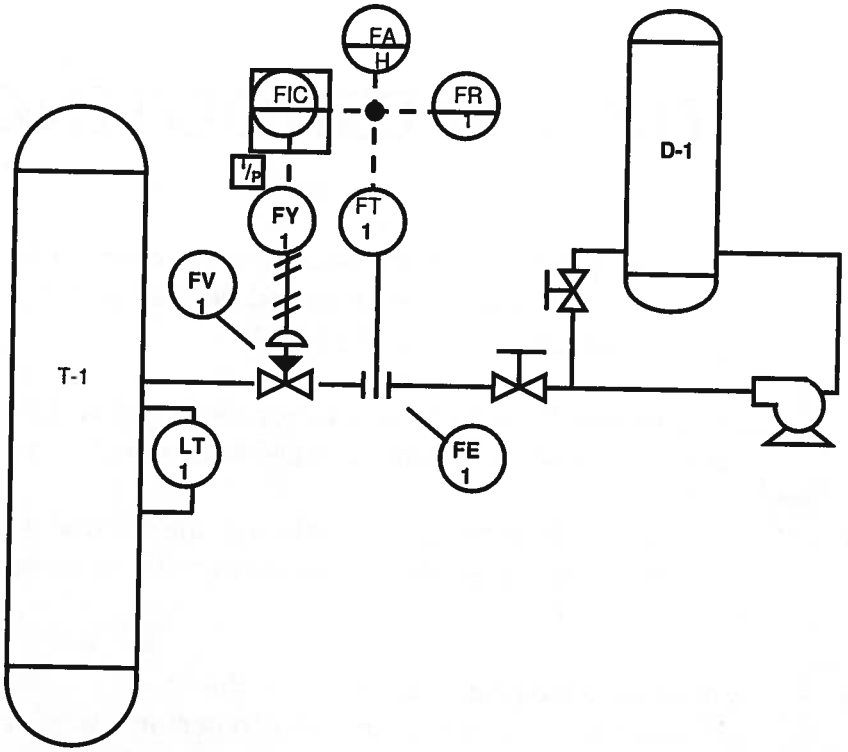
A. feedback feedwater control.
B. single element feedwater control.
C. two element feedwater control.
D. three element feedwater control.

31. In this "bubbler" level control application, FI-1:



A. sets LIC-1 setpoint.
B. controls purge gas flow.
C. controls gas pressure.
D. indicates purge gas flow.

32. This tower feed flow control would not work well as part of an interactive control system because it:



A. is a pneumatic loop.
B. is not connected to the level transmitter, LT-1.
C. does not have square root extraction.
D. is dependent on the level in D-1.

33. Which of the following types of actuators responds to a pneumatic signal?

A. solenoid
B. motor
C. diaphragm
D. electromagnetic valve

34. Changing the point of origin for input data, and the output signal destination is easy with a Distributed Control System because many changes can be implemented by:

A. relocating wire jumpers in a junction box.
B. changing the pneumatic tubing connections.
C. reassigning inputs/outputs by changing loop configuration.
D. installing new transmitters and final control elements.

PERFORMANCE DOMAIN 3

TROUBLESHOOTING

Task 1 Examine all pertinent documentation in order to identify the proper operation of the control system and devices and to provide a baseline against which to evaluate the problem.

Task 2 Identify any discrepancies between observed and proper operation of the control system in order to determine whether a problem exists.

Task 3 Determine which device or devices of the control system could be causing the observed discrepancies in order to formulate a plan of corrective action.

Task 4 Isolate each suspected device from the control system by following established site procedures in order to perform troubleshooting in a safe manner.

Task 5 Exercise each device across calibrated range in order to determine proper operation.

Task 6 Perform corrective action, if necessary, according to site procedures in order to resolve the problem.

Task 7 Complete all documentation as required in order to update all records.

1. What is a thermowell?

   A. protective tube
   B. type of manometer
   C. expandable bulb
   D. reservoir

2. The body of a valve may contain packing material that prevents leakage:

   A. along the valve stem.
   B. between the seat and the disk.
   C. between the bonnet and the body.
   D. between the inlet and outlet ports of the valve.

3. Which type of flowmeter does a rotor mounted axially within a pipe between a set of bearings describe?

   A. magnetic
   B. turbine
   C. water meter
   D. venturi tube

4. Which type of flowmeter measures flow by measuring volume directly?

   A. coriolis
   B. magnetic
   C. positive displacement
   D. differential pressure

5. What occurs if the temperature of the thermocouple measuring junction is lower than the reference junction?

   A. There is no emf output.
   B. The output voltage polarity is reversed.
   C. The polarity stays the same but voltage increases.
   D. The emf remains the same when temperature changes.

6. A ground loop on shielded cable can be avoided by:

   A. attaching a safety ground to the instrument casing.
   B. grounding the shield at only one end of the circuit.
   C. attaching a safety ground to the control panel.
   D. grounding the shield at both ends of the cable.

7. A meter reading of 1 ohm between an ungrounded signal lead and the earth ground indicates a problem of:

   A. a lead having an unwanted connection to ground.
   B. an open circuit from the lead to ground.
   C. an open circuit from the lead to the casing.
   D. none of the above

8. What kind of wires in a process control loop would most likely be shielded cables?

   A. power wires
   B. ground wires
   C. signal wires
   D. control wires

9. A device used to change an analog signal to a digital signal is most often called a:

   A. transducer.
   B. signal conditioner.
   C. transmitter.
   D. converter.

10. A breadboarding area is:

    A. a data routing circuit.
    B. used to design or test electronic circuits.
    C. a code-converting circuit.
    D. a temporary binary storage area.

11. A 4-20 mA electronic transmitter has an input range of 50-330 degrees F. If the output is 13.77 mA, what is the indicated temperature in degrees F?

    A. 173
    B. 221
    C. 266
    D. 302

12. If the range of a 3-15 psi pneumatic transmitter is 50-350 degrees F, what is the output signal value, in psi, at 198 degrees F?

A. 5.7
B. 7.2
C. 8.9
D. 12.6

13. When the percentage of flow through a valve equals the percentage of plug movement, a valve has:

A. linear flow characteristic.
B. equal percentage flow characteristic.
C. quick opening flow characteristic.
D. curved flow characteristic.

14. Which of the following parts of a globe valve serves the same purpose as the disk in a butterfly valve?

A. seat
B. plug
C. packing rings
D. packing flange

15. The process of grinding the plug and seat of a valve is called:

A. characterizing.
B. linearizing.
C. lapping.
D. calibrating.

16. The dead band of an instrument is:

A. the band that prevents the instrument from being tampered with by unauthorized persons.
B. the range of valves for which the instrument gives inaccurate readings.
C. the size of the instrument indicated by the divisions on the scale of the instrument.
D. the range that an input signal may be changed on reversal of direction without an observable change in the output.

17. A continuity check should be made prior to hook-up of a field transmitter to the control room so that the technician:

A. can assure that the transmitter is being hooked-up to the proper indicator/controller.
B. can make sure that the proper voltages are present.
C. can make sure that he continue hook-up on a particular transmitter.
D. knows that he may connect another transmitter to the loop circuit.

18. Impulse lines are used with:

A. Pressure meters.
B. magnetic flowmeters.
C. positive displacement meters.
D. open channel meters.

19. Which meter does not use an obstruction?

A. head meter
B. magnetic meter
C. open channel meter
D. turbine meter

20. The Coriolis effect is associated with the:

A. head meter.
B. mass flowmeter.
C. positive displacement meter.
D. turbine meter.

21. A beveled orifice should form an angle of:

A. not less than 30 degrees to the axis of the pipe.
B. not more than 30 degrees to the axis of the pipe.
C. not less than 45 degrees to the axis of the pipe.
D. not more than 45 degrees to the axis of the pipe.

22. Which orifice plate is used to eliminate damming of material at the top or bottom of the pipe?

A. concentric
B. round-edged
C. quadrant-edged
D. segmental

23. The most common pressure tap is the:

A. corner.
B. flange.
C. pipe.
D. vena contracta.

24. One of the most important considerations of the installation of an electromagnetic flowmeter is:

A. the Reynolds Number of the fluid.
B. the use of a straightening vane.
C. bonding the flowmeter to the piping.
D. observing the correct polarity.

25. Control valves which have a plug positioned by a stem which slides through a packing gland are often referred to as:

A. rotary motion control valves.
B. linear motion control valves.
C. ball control valves.
D. motorized control valves.

26. The control valve that produces an inherent equal percentage of increase, or decrease, over the existing flow when the plug is repositioned has a (an) _____ characteristic under constant pressure drop conditions.

A. modified parabolic
B. equal percentage
C. linear
D. quick opening

27. If a control valve will pass 25% of its full capacity when it is 25% open, 50% of its capacity when 50% open, and 75% when 75% open, the inherent characteristic is:

A. quick opening.
B. equal percentage.
C. throttling.
D. linear.

28. If the pressure drop (delta P) across a valve increases, without a change in valve position, the flow of fluid through the valve will normally:

A. decrease.
B. increase.
C. remain the same.
D. change direction.

29. A piston actuator:

A. seldom has a pneumatic positioner.
B. can only be actuated by hydraulic pressure above 100 psig.
C. cannot be made fail safe with out electric limit switches.
D. can be used if a long stroke or high actuator pressure is required.

30. An instrument calibration error is reported. Your analysis of the calibration plot (Fig. 3-1) determines that _____, and the instrument does, in fact, need recalibration.

A. there is a zero error
B. there is a span error
C. there is both a zero and span error
D. there is a linearity error

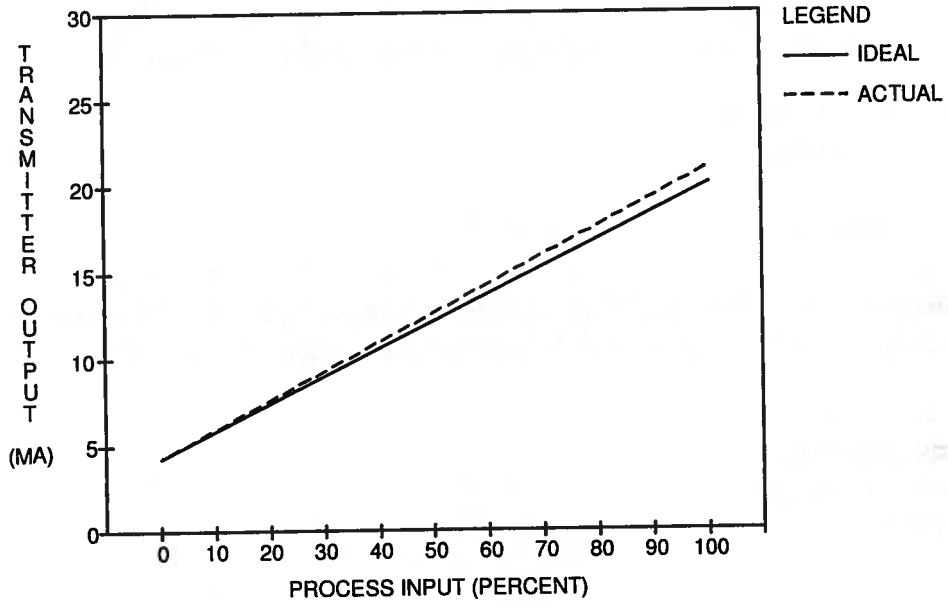


Figure 3-1

31. Which of the following is not necessary when zeroing or calibrating a magnetic flowmeter?

A. calibration factor for the flow tube assembly
B. flow tube full of process material
C. proper grounding
D. warm-up period for the fluid in the system

32. A dp cell is mounted 3 ft below the zero level in an open tank. For a true level reading, the transmitter output must be:

A. calibrated for elevated zero.
B. calibrated for span.
C. calibrated for true zero.
D. calibrated for suppressed zero.

33. The standard range for pneumatically transmitted signals is:

A. 3-15 psig.
B. 5-20 psig.
C. 1-10 psig.
D. 4-02 psig.

34. Approximately what is the pressure, in psi, at the bottom of a 20 ft tank filled to 100% of capacity with a fluid that has a specific gravity of 1.5?

A. 8.7
B. 12.99
C. 5
D. 35

35. A process variable which is to be maintained at some desired value (temperature, pressure, level, flow), by means of manipulating another process variable, is the:

A. manipulated variable.
B. measured variable.
C. controlled variable.
D. disturbance variable.

36. A plant has a water tank mounted on top of an 80 foot platform. The tank is 30 ft. high. What is the level of water in the tank if a differential pressure transmitter mounted on the bottom of the tank reads 277 inches of water?

A. 10 feet
B. 17 feet
C. 23 feet
D. 30 feet

37. The pressure at the bottom of a 20 foot tank completely filled with a fluid that has a specific gravity of 1.75 will be:

A. 11.428 psi.
B. 15.15 psi.
C. 21.75 psi.
D. 35 psi.

38. Pneumatic actuators:

A. cannot be used in hazardous or high humidity atmospheres.
B. can be made fail safe.
C. do not have sufficient speed or power for most applications.
D. are too complicated and unreliable for many applications.

39. A hollow, metal cylinder which guides the movement of the plug and holds the seat ring in place is found in a:

A. seat.
B. butterfly valve.
C. cage valve.
D. ball.

40. Per ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification* , this is a (an) _____ Instrument Line Symbol.

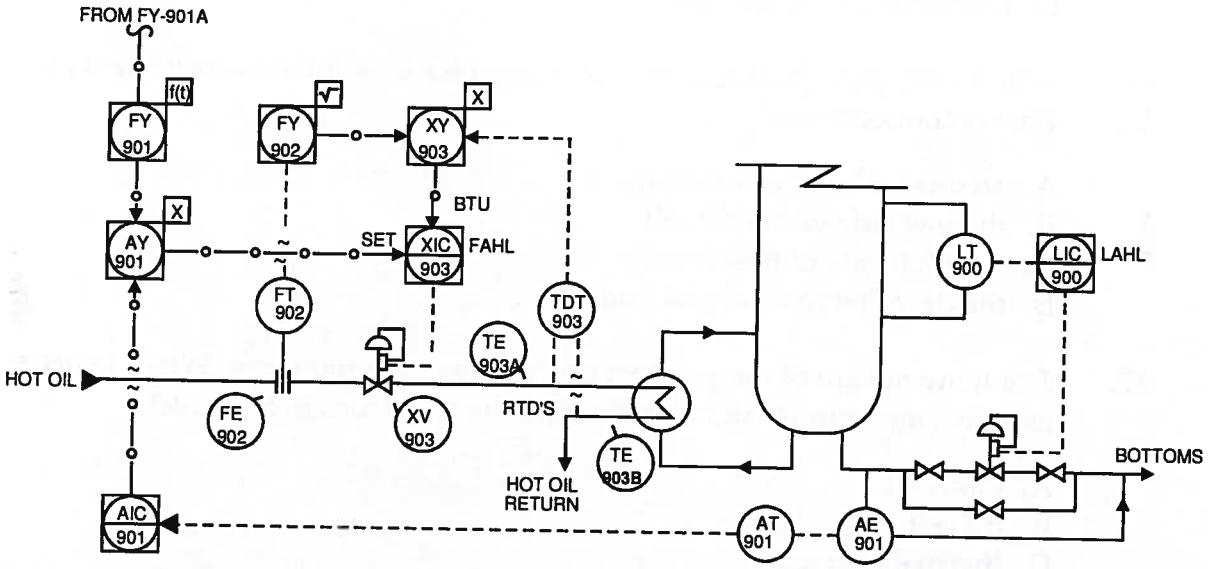
A. Electric
B. Pneumatic
C. Hydraulic
D. Sonic

41. The line symbol from ISA-5.1-1984 (R1992) -- *Instrument Symbols and Identification*  identifies a(an):

A. mechanical link.
B. electrical signal.
C. pneumatic signal.
D. hydraulic signal.

42. In this drawing, FY - 902 is a(an) _____ function.

A. analysis
B. multiplication
C. square root
D. summing



43. The term valve "trim" refers to the:

A. pneumatic actuator.
B. stem, plug, and seat ring.
C. orientation in a process installation.
D. body style and material of construction.

44. In a bubbler level measuring system with the open end of the purge pipe at the bottom of the tank, the air pressure in the pipe is _____ the pressure exerted by the liquid at the bottom of the tank.

A. less than
B. equal to
C. twice
D. three times

45. As the temperature rises in a closed vessel, the pressure:

A. increases.
B. decreases.
C. stays the same.
D. increases then decreases.

46. When a dip pipe (bubbler) system is used for tank level measurement, the dip pipe is connected to:

A. an unregulated gas supply.
B. the low side of the dP cell.
C. the high side of the dP cell.
D. the dP cell output signal line.

47. The temperature of the product in a storage tank increases. What device will still give an accurate signal indicating the true mass in the tank?

A. load cell
B. dP cell
C. thermal element
D. float

48. In which of the following does increasing signal pressure cause the actuator stem to retract and close a valve?

A. direct action, air-to-close
B. direct action, air-to-open
C. reverse action, air-to-close
D. reverse action, air-to-open

49. A field mounted pressure device that changes a process variable measurement into a signal that can be used by a receiving instrument located in a control room is usually called a:

A. D/P transducer.
B. Signal Conditioner.
C. D/P transmitter.
D. Signal Converter.

50. The pressure taps in a standard set of orifice flanges are located _____ upstream from the leading edge of the orifice, and _____ downstream from the trailing edge of the orifice plate.

A. 8 inches, 5 inches
B. 5 inches, 8 inches
C. 30 inches, 10 inches
D. 1 inch, 1 inch

51. A bench test of a spring and diaphragm actuator with a positioner shows that full stem travel is not obtained even when full supply pressure is available to the positioner. Which of the following conditions is the most likely cause of the problem?

A. The actuator diaphragm area is too large.
B. The stem is too thick.
C. A stem travel stop is out of adjustment.
D. The spring is too weak.

52. The oil within the variable capacitor of a capacitance-type differential pressure transmitter is separated from the process by:

A. a sensing diaphragm.
B. capacitor plates.
C. isolating diaphragms.
D. bellows.

53. The leakage of a metal-seated valve may be reduced by a process called:

A. lapping.
B. calibrating.
C. linearizing.
D. characterizing.

54. The results of an orifice plate installed with the bevel on the wrong side is:

A. damage to the plate.
B. erratic meter readings.
C. high meter readings.
D. low meter readings.

55. When selecting calibration equipment, a CST must verify that the:

A. calibration equipment is more accurate than the device to be calibrated.
B. device to be calibrated is more accurate than the test equipment.
C. controller positioning the control valve is accurate.
D. process is diverted from the device to be calibrated.

56. The output signal from a digital control system can NOT be in response to _____ input.

A. one
B. more than one
C. less than one
D. an analog

PERFORMANCE DOMAIN 4

START-UP

Task 1 Examine all pertinent documentation in order to verify completion of all control systems that are required for start-up.

Task 2 Implement applicable safety practices to be followed in order to ensure that all personnel are informed of the possible hazards during start-up.

Task 3 Coordinate activities with all personnel involved in order to ensure that all personnel are aware of their responsibilities during start-up.

Task 4 Implement start-up according to site procedures in order to ensure that the system is operating properly.

Task 5 Assist site personnel to identify and correct problems that arise during start-up in order to commission the system.

Task 6 Complete all documentation as required in order to update all records.

1. The _____ variable is the part of a process which is adjusted to close the gap between the set point and the value of the controlled variable.

   A. manipulated
   B. measured
   C. controlled
   D. disturbance

2. The input to a controller that can be manually set, automatically set, or programmed to determine the desired value of the controlled variable is called the:

   A. measurement signal.
   B. set point.
   C. controlled variable.
   D. output signal.

3. An undesired change in a variable which tends to adversely affect the value of a controlled variable, is normally called a (an):

   A. process lag.
   B. measurement error.
   C. disturbance.
   D. error signal.

4. In a home heating system, the room temperature is the:

   A. manipulated variable.
   B. setpoint.
   C. controlled variable.
   D. disturbance variable.

5. The integral action which is part of a PID algorithm is often called:

   A. rate.
   B. reset.
   C. gain.
   D. insert.

6. In instrumentation, the term "loop" usually defines:

   A. a circle in the instrument signal lines to help minimize effects of vibration.
   B. a circle in the instrument piping to reduce water hammer.
   C. a combination of two or more inter-connected instruments.
   D. an internal RC filtering circuit to reduce electrical noise.

7. The term "primary element" usually refers to:

   A. PID control action.
   B. pneumatic control valves.
   C. transmitters and transducers.
   D. detectors or sensors.

8. Most pneumatic controllers:

   A. produce a 6 to 30 psig output.
   B. have some type of baffle nozzle.
   C. are PID controllers.
   D. can accept a 4 to 20 mA input.

9. A pneumatic controllers output is a function of the spacing between a baffle and a nozzle. A change in the baffle nozzle relationship produces an increase or decrease in back pressure that is sensed by the relay. This baffle/nozzle relationship is determined by the:

   A. relay.
   B. output signal.
   C. input signal.
   D. supply air signal.

10. Dew point in an instrument air supply system is usually measured:

    A. after the air leaves the separator.
    B. after the air leaves the dryer.
    C. before the air enters the dryer.
    D. before the air enters the receiver.

11. An analog multi-meter may be used to measure:

    A. frequency.
    B. cycles.
    C. voltage.
    D. pressure.

12. The time it takes for an instrument to produce a change in output in response to a change in input is called:

A. reaction time.
B. dead time.
C. lead time.
D. deviation time.

13. As the temperature of a liquid increases, its density:

A. increases.
B. doubles.
C. remains the same.
D. decreases.

14. The level measurement installation (Fig. 4-1) infers level by measuring:

A. variable capacitance.
B. hydrostatic head.
C. positive displacement.
D. thermal conductivity.

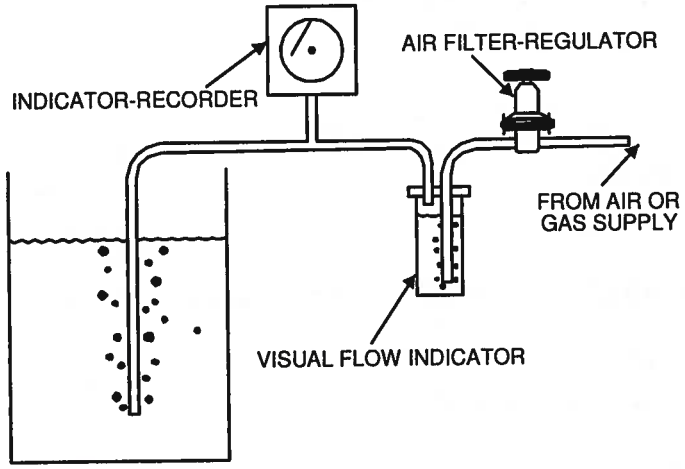


Figure 4-1

15. In a two stage phenomenon of liquid flow, the first stage is the formation of voids or cavities within the liquid system. The second stage is the collapse or implosion of these cavities back into an all-liquid state called:

A. cavitation.
B. corrosion.
C. erosion.
D. flashing.

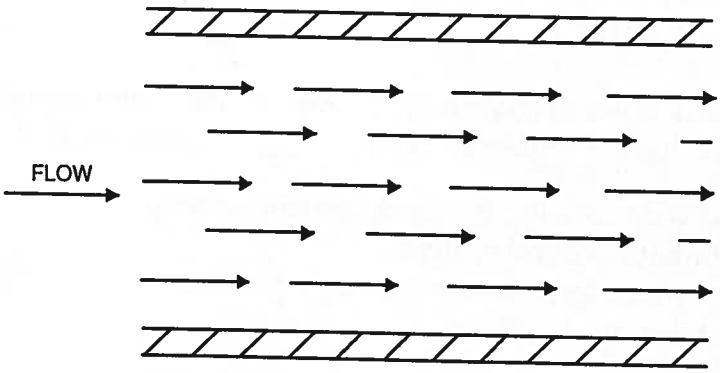
16. The most important function in safety belongs to:

A. the safety professional.
B. the owner or president of the company.
C. supervisors.
D. OSHA.

17. A new feedback cam is required for a positioner. What specific documentation is required to change cam A to cam B in the positioner?

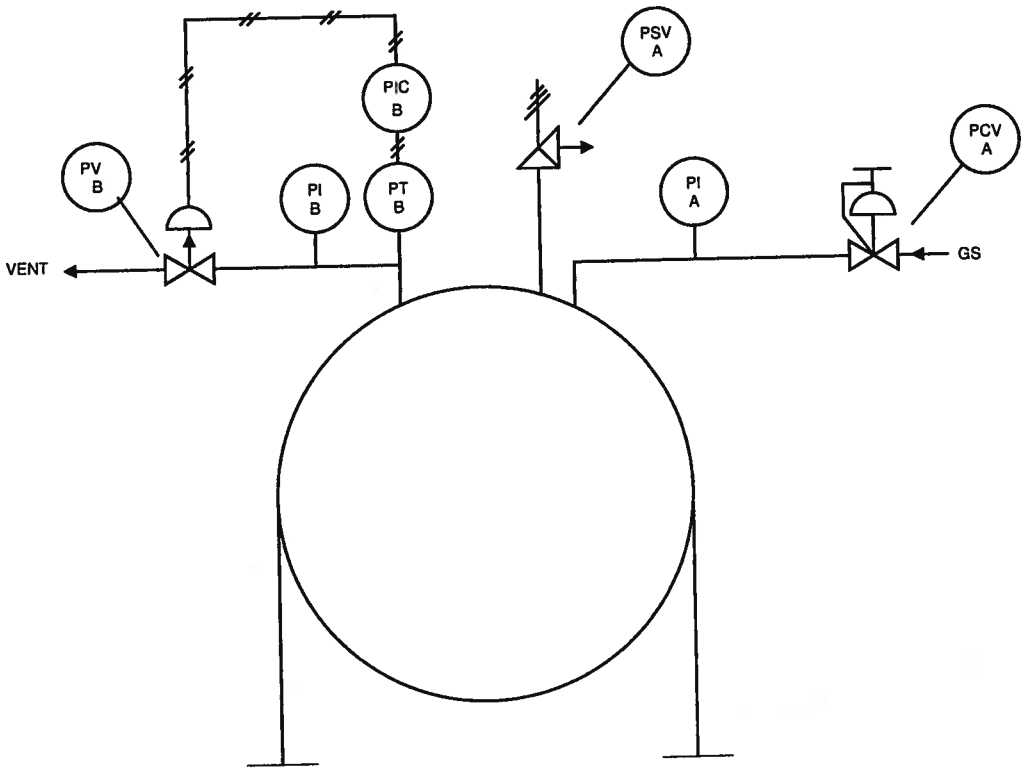
A. Operation and maintenance manual
B. Instrument index and instruction book
C. Calibration procedures
D. Positioner standard

18. A representation of _____ flow is shown below.



A. turbulent
B. laminar
C. erratic
D. transition

19. The pneumatic control valve in this storage tank pressure control system:



A. will fail open.
B. will fail locked.
C. will fail closed.
D. will never fail.

20. It is much easier to coordinate process controls and implement complex interactive control requirements with _____ than with other systems.

A. hard wired analog feedback control systems
B. pneumatic control systems
C. hard wired logic function (relay) systems
D. digital control systems

21. _____ control is often used when a specific blend of two different materials is required.

A. Ratio
B. Cascade
C. Feedforward
D. Feedback

→ 22. Which of the following requires no "power" source?

A. hydraulic load cell
B. bonded strain gage
C. hydrostatic head transmitter
D. rotating paddle

23. Limit switch cams are usually found in:

A. solenoid actuators.
B. diaphragm actuators.
C. piston actuators.
D. electric motor actuators.

24. Which of the following types of actuators offers high thrust as a result of being able to use high air pressures?

A. rolling diaphragm
B. piston
C. spring and diaphragm
D. electromechanical

25. What is the purpose of internal cache memory in a computer system?

A. store, temporarily, the results of each calculation
B. retain program instructions until they are used
C. speed up the performance of a CD-ROM drive
D. provide back-up memory for RAM

26. In which register for a standard microprocessor is the result of arithmetical and logical operations stored?

A. Accumulator
B. Address
C. General purpose
D. Processor status

27. Which of the following does NOT apply to asynchronous serial transmission?

A. Data characters can be sent with any length interval between them.
B. Characters are framed by stop and start bits.
C. Clocking and timing techniques are used for transmission.
D. Message coding provides coordinates data transfer.

28. The status given to the parity bit for a selected parity mode is based on the number of ones in the:

A. bits in the data word, including parity.
B. bits in the data word, parity and the stop bit.
C. bits of the data word.
D. start bit and bits in the data word.

Solutions to Questions

Domain 1/ Calibration:

1C, 2D, 3D, 4C, 5B, 6A, 7B, 8C, 9B, 10A, 11C, 12C, 13C, 14A, 15C, 16B, 17C, 18D, 19D, 20D, 21A, 22D, 23B, 24D, 25A, 26A, 27D, 28D, 29A, 30D, 31C, 32A, 33C, 34C, 35D, 36B, 37C, 38A, 39A, 40D, 41A, 42D, 43B, 44B, 45A, 46C, 47A, 48D, 49C, 50B, 51C, 52C, 53C, 54B, 55B, 56D, 57C, 58D, 59D

Domain 2/ Loop Checking:

1D, 2B, 3B, 4B, 5D, 6B, 7B, 8C, 9B, 10D, 11B, 12D, 13B, 14A, 15C, 16D, 17A, 18D, 19C, 20A, 21A, 22B, 23B, 24B, 25D, 26B, 27D, 28D, 29B, 30D, 31D, 32C, 33C, 34C

Domain 3/ Troubleshooting:

1A, 2A, 3B, 4C, 5B, 6B, 7A, 8C, 9D, 10B, 11B, 12C, 13A, 14B, 15C, 16D, 17A, 18A, 19B, 20B, 21C, 22D, 23B, 24C, 25B, 26B, 27D, 28B, 29D, 30B, 31D, 32D, 33A, 34B, 35C, 36C, 37B, 38B, 39C, 40A, 41C, 42C, 43B, 44B, 45A, 46C, 47A, 48C, 49C, 50D, 51C, 52C, 53A, 54D, 55A, 56C

Domain 4/ Start-Up:

1A, 2B, 3C, 4C, 5B, 6C, 7D, 8B, 9C, 10B, 11C, 12B, 13D, 14B, 15A, 16C, 17A, 18B, 19A, 20D, 21A, 22A, 23D, 24B, 25B, 26A, 27C, 28C

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2 Loop Checking
3 Troubleshooting
4 Start-Up

Level 2
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2 Loop Checking
3 Troubleshooting
4 Start-Up
5 Maintenance/Repair
6 Project Organization

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2 Loop Checking
3 Troubleshooting
4 Start-Up
5 Maintenance/Repair
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CCST® Study Guide Level II	X	X	X	X	X	X	
CCST® Study Guide Level III	X	X	X	X	X	X	X
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Calibration: A Technician's Guide	X						
Condensed Handbook of Measurement & Control, 2nd Edition	X	X	X	X			
Control System Documentation: Applying Symbols and Identification		X		X	X		
Basic Math for Process Control	X	X	X	X			
Control System Safety Evaluation and Reliability, 2nd Edition			X				X
Electrical Instruments in Hazardous Locations, 4th Edition			X				X
Fundamentals of Process Control Theory, 3rd Edition		X					
Good Tuning: A Pocket Guide	X	X					
Industrial Ethernet				X			
ISA Handbook of Measurement Equations and Tables	X						

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	1	2	3	4	5	6	7
Loop Checking: A Technician's Guide		X					
Maintenance of Instruments & Systems, 2nd Edition	X	X	X		X	X	X
Measurement and Control Basics, 3rd Edition	X			X			
Preventive Maintenance, 3rd Edition					X	X	
Process Control: A Primer for the Non-specialist and the Newcomer, 2nd Edition	X	X					
Programmable Controllers, 3rd Edition			X	X			
Reading a P&ID		X		X	X		
Safety Instrumented Systems: Design, Analysis and Justification, 2nd Edition				X			
Start-Up: A Technician's Guide				X			
Successful Instrumentation and Control Systems Design	X	X	X	X	X	X	X
Troubleshooting: A Technician's Guide			X				
Tuning of Industrial Control Systems, 2nd Edition		X					
Understanding Distributed Processor Control Systems for Control			X				
Wireless Networks			X	X			
CD & ONLINE COURSES							
Feedback Control		X					
Process Control Modes		X					
Process Characteristics	X			X			
Process Variables				X			
Instrumentation Symbols					X		
Instrument Loop Diagrams					X		
Piping and Instrumentation Drawings					X		
Mechanical Connections		X					
Electrical Connections		X					
Temperature 1: Thermometers and Thermocouples				X			
Temperature 2: Resistance and Radiation Devices				X			
Pressure 1: Manometers and Gages				X			
Pressure 2: Indicators and Transmitters				X			
Level 1: Level Measurement and Gages				X			
Level 2: Level Indicators and Transmitters				X			
Flow 1: Flow Measurement Overview				X			
Flow 2: Flow Sensors				X			
Primary Calibration Standards	X						
Pneumatic Test Equipment	X						
Electronic Test Equipment	X						
Oscilloscopes	X						
Instrumentation Errors	X						
Instrumentation Calibration	X						

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	1	2	3	4	5	6	7
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Configuring& Changing Control Strategy							X
Principles of Continuous Control				X			
Applications of Heat Exchanger Control			X				
Applications of Distillation Control							X
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Control Valves & Actuators: Types & Design			X			X	
SOFTWARE							
PC-ControLAB		X					
STANDARDS							
ISA5.1-1984 (R1992)		X	X	X	X		
ISA20-1981					X	X	
ISATR20.00.01-2001					X	X	
TRAINING COURSES							
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EI30: Sizing, Selecting, and Applying Process Control Valves			X	X	X		X
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ES35: Evaluating Control System Safety and Reliability			X	X	X	X	
FG07: Introduction to Industrial Automation and Control	X	X		X			X
SP15: Understanding and Applying Motor Control And Drives		X	X	X	X		
TC05: Understanding and Tuning Control Loops		X	X				
TC10: Troubleshooting Instrumentation and Control Systems	X	X	X				
TC35P: PLC Automation: PLC Structure, Programming, Installation, and Maintenance		X	X	X	X		
TI06: Maintaining Pneumatic Components in Measurement and Control	X	X	X	X			X
TI15: Understanding Electrical Systems	X	X					X
TI20: Industrial Electronics	X	X					X
TI21: Grounding and Noise Considerations for Control Equipment and Computers			X	X	X		
TI25: Installing, Calibrating, and Maintaining Electronic Instruments	X	X	X	X	X	X	X
TS05: Industrial Data Communications, Part I			X				
TS10: Industrial Data Communications, Part II			X				
TS15: Smart Devices, Advanced Operations	X	X	X	X	X		X
TS00: Certified Control Systems Technician® (CCST®) Review Course	X	X	X	X			

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